

Day 22 – AI Multi-Agent Systems using AutoGen & Python

Project 1:

AI Career Counseling Multi-Agent System

The image shows two terminal windows side-by-side, displaying the execution of a Python script for an AI Career Counseling Multi-Agent System.

Left Terminal Window:

```
PS C:\Users\abc\OneDrive\Desktop\auto gen day 21\career_counseling_agent> & C:\Users\abc\AppData\Local\Microsoft\WindowsApps\python3.12.exe "c:\Users\abc\OneDrive\Desktop\auto gen day 21\career_counseling_agent\main.py"

=====
AI CAREER COUNSELING MULTI-AGENT SYSTEM
=====

Enter Student Name: Arashpreet Kaur
Enter Qualification: B.Tech Computer Science Engineering (3rd Year)
Enter Current Skills (comma separated): Python, C++, HTML, CSS, SQL, Git, GitHub, Linux, VS Code
Enter Interests: Artificial Intelligence, Machine Learning, Web Development, Cybersecurity, Automation
Enter Career Goal: AI Engineer

=====
Initializing AI Career Counseling Multi-Agent System...
[INFO] Loading Career Analyst Agent...
[SUCCESS] Career Analyst Agent Initialized.

[INFO] Loading Skill Assessment Agent...
[SUCCESS] Skill Assessment Agent Initialized.

[INFO] Loading Learning Path Agent...
[SUCCESS] Learning Path Agent Initialized.

[INFO] Loading Resource Recommendation Agent...
[SUCCESS] Resource Recommendation Agent Initialized.

[INFO] Loading AI Roadmap Planner Agent...
[SUCCESS] AI Roadmap Planner Initialized.

[INFO] Loading Progress Tracker Agent...
[SUCCESS] Progress Tracker Ready.

=====
ANALYZING STUDENT PROFILE
=====
Student Name      : Arashpreet Kaur
Qualification     : B.Tech Computer Science Engineering (3rd Year)
Career Goal      : AI Engineer
```

Right Terminal Window:

```
=====
Learning Path Agent
=====
Recommended Learning Roadmap

Phase 1
✓ Master Python
✓ Data Structures & Algorithms
✓ SQL & Database Design

Phase 2
✓ NumPy
✓ Pandas
✓ Data Visualization
✓ Statistics for AI

Phase 3
✓ Machine Learning
✓ Deep Learning
✓ Computer Vision
✓ Natural Language Processing

Phase 4
✓ FastAPI
✓ Flask
✓ Docker
✓ GitHub Portfolio

Phase 5
✓ Build Real-world AI Projects
✓ Deploy AI Applications
✓ Participate in Hackathons
✓ Apply for AI Internships

=====
Resource Recommendation Agent
=====
Recommended Platforms
• Coursera
• Udemy
• freeCodeCamp
• Kaggle
• Google AI
• Microsoft Learn
• Hugging Face
• OpenCV Documentation

Recommended Books
• Hands-On Machine Learning
• Python Crash Course
• Deep Learning with Python
• Introduction to Machine Learning with Python

=====
AI Roadmap Planner
=====
Next 30 Days
✓ Strengthen Python
✓ Practice DSA Daily
✓ Complete NumPy & Pandas

Next 60 Days
✓ Learn Machine Learning
✓ Build 2 AI Projects

Next 90 Days
✓ Learn Deep Learning
✓ Build Portfolio
✓ Deploy AI Applications
✓ Apply for Internships

=====
Progress Tracker
=====
Career Readiness Score : 88%

Programming          ██████████ 95%
Problem Solving      ██████████ 90%
Machine Learning     ██████████ 45%
Deep Learning        ██████████ 35%
Portfolio            ██████████ 70%
Communication         ██████████ 80%

(Output continues in next part...)
```

Project 2

AI Multi-Agent Project Planning System

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - [ ] [X]
(venv) PS C:\Users\abc\OneDrive\Desktop\auto gen day 22\multi_agent_project_planner> python main.py
=====
AI MULTI-AGENT PROJECT PLANNING SYSTEM
=====
Enter your project idea (or 'exit' to quit):
Build a Student Attendance Management System

Generating project plan using multiple AI agents... Please wait!

🔍 1. REQUIREMENT ANALYST AGENT
-----
Status: Completed

🔧 2. TECHNICAL ARCHITECT AGENT
-----
Status: Completed

📁 3. PROJECT MANAGER AGENT
-----
Status: Completed

🛡️ 4. REVIEWER AGENT
-----
Status: Completed - Suggestions Generated

📄 5. FINAL REPORT AGENT
-----
Status: Completed - Final Plan Ready

📋 FINAL PROJECT PLAN GENERATED SUCCESSFULLY!
=====
Total Agents Executed : 5/5
Overall Quality Score : 9.2 / 10
Report Saved To : final_project_plan.md
=====
Thank you for using the AI Multi-Agent Project Planning System!
(venv) PS C:\Users\abc\OneDrive\Desktop\auto gen day 22\multi_agent_project_planner>
```